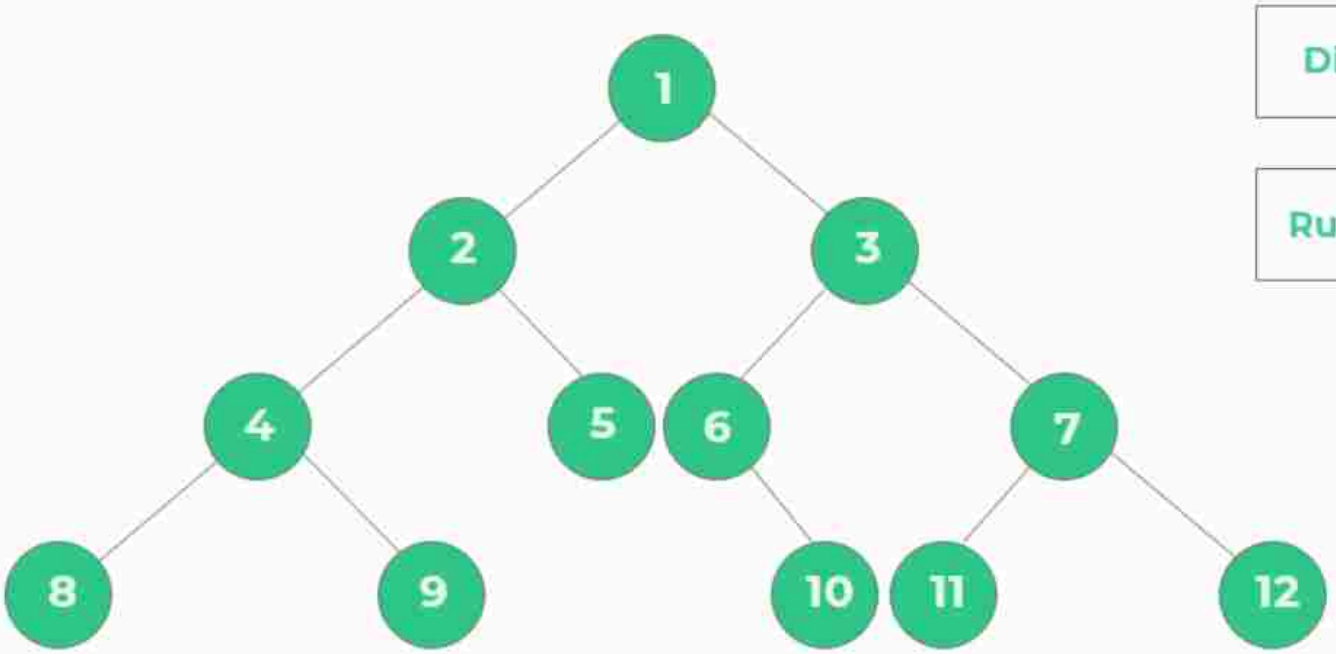
DS	**Session**	Tree 1	**Question Information**	Level 1 Challenge 61

Question description

Jegan Sir students were chatting and playing quite loudly on the last day of the year, celebrating the end of the academic session. Jegan sir was harshly chastised by the college's principal. But, instead of becoming enraged, he attempted to engage everyone in a different task.

So Jegan sir gave his students to solve the task such that, you have to perform pre-order tree traversal in Binary search tree.



Direction : Anti Clock

Rule : CLR (Center-Left-Right)

PreOrder:

1 2 4 8 9 5 3 6 10 7 11 12

Constraints:

0<size <100

0<data<1000

Input format:

First line indicates the size of the tree

Second line indicates the elements of the tree

Output Format:

single line indicates the pre-order list of the given value

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
7
200 120 80 160 280 240 320
```

EXPECTED OUTPUT

```
200 120 80 160 280 240 320
```

Test Case 2

INPUT (STDIN)

```
9
34 67 10 90 410 810 40 20 60
```

EXPECTED OUTPUT

```
34 10 20 67 40 60 90 410 810
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void preorder(struct node* root)
```

Test Case 2

KEYWORD

```
struct node* newNode(int item)
```

Test Case 3

KEYWORD

```
struct node
```

Test Case 4

KEYWORD

```
struct node *left,*right;
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
2
```

Test Case 2

TOKEN COUNT

```
280
```

Test Case 3

NLOC

```
50
```

Code

solution.c

```c
#include <stdio.h>
#include <stdlib.h>

struct node {
    int data;
    struct node *left, *right;
};

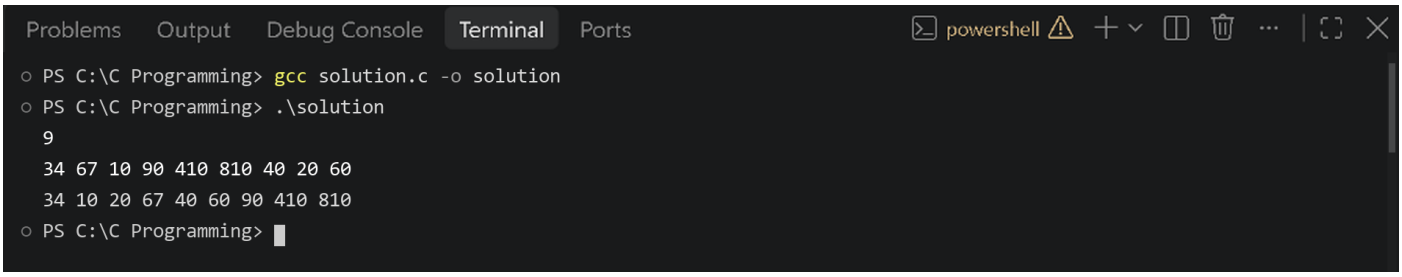
struct node* newNode(int item) {
    struct node* temp = (struct node*)malloc(sizeof(struct node));
    temp->data = item;
    temp->left = temp->right = NULL;
    return temp;
}

struct node* insert(struct node* node, int data) {
    if (node == NULL) return newNode(data);
    if (data < node->data) node->left = insert(node->left, data);
    else if (data > node->data) node->right = insert(node->right, data);
    return node;
}

void preorder(struct node* root) {
    if (root != NULL) {
        printf("%d ", root->data);
        preorder(root->left);
        preorder(root->right);
    }
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    struct node* root = NULL;
    for (int i = 0; i < n; i++) {
        int x; scanf("%d", &x);
        root = insert(root, x);
    }
    preorder(root);
    printf("\n");
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
9
34 67 10 90 410 810 40 20 60
34 10 20 67 40 60 90 410 810
PS C:\C Programming>